

Arkapriyo Hore

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Education

Indian Statistical Institute B.Math(Hons.)

Bengaluru, Karnataka
July, 2023 – Present

Relevant Coursework: Real Analysis; Linear Algebra; Probability Theory; Elementary Number Theory; Discrete Mathematics; Group Theory; Multivariate Analysis; Rings and Modules; Topology; Ordinary Differential Equations; Field and Galois theory; Function Spaces; Algebraic Topology; Representation theory of finite groups

Delhi Public School Ruby Park, Kolkata Passed AISSCE with 84%

Kolkata, West Bengal
2020 – 2022

St. Mary's School, Malda Passed ICSE with 97.2%

Malda, West Bengal
2009-2020

Projects

Galois Cohomology and Brauer-Severi Variety *Prof. Debanjana Kundu (University of Regina)*

June, 2025 - July, 2025

For the project I studied the books **Local Fields** by Jean- Pierre Serre and **Central Simple Algebras and Galois Cohomology** by Philippe Gille & Tamás Szamuely. During the project I covered the concepts of Non-abelian Galois Cohomology, Cohomological Dimension, Brauer Group and its connection with central simple algebras and algebro-geometric aspects of Brauer-Severi Varieties.

Visiting Student's Research Program *Prof. Sandeep Varma (School of Mathematics, TIFR Mumbai)*

May, 2025 - June, 2025

For the project I studied the book **Local Fields** by Jean- Pierre Serre. During the project I covered the concepts of Group Cohomology, Class Formations, and thereby using it to prove theorems in Local Class Field Theory and the Local Kronecker–Weber theorem following which I wrote a brief report titled *A Journey through Local Class Field Theory*.

Sheaf Theory *Prof. Suresh Nayak (ISI Bangalore)*

November, 2024 - December, 2024

For the project I studied the book **The Rising Sea Foundations of Algebraic Geometry** by Ravi Vakil. During the project I covered concepts of Homological Algebra, definition of sheaf and presheaf and morphism between them, sheafification and sheaves of abelian groups.

Arithmetic Theory of Binary Quadratic Forms *Prof. Satadal Ganguly (ISI Kolkata)*

May, 2024 - July, 2024

For the project I studied the book **Binary Quadratic Forms** by Duncan A. Buell. During the project I covered the concepts of Gauss composition, Class Groups, Ideal Class Groups following which I gave a presentation titled *Relation between Class groups of Binary Quadratic Forms and Ideal Class Groups in Quadratic Fields*.

Introduction to Number Fields and Number Rings *Prof. Ramdin Mawia (ISI Bangalore)*

February, 2024 - April, 2024

For the project I studied the book **Number Fields** by Daniel A. Marcus. During the project I covered the basics of Commutative Algebra, Galois Theory and Algebraic Number Theory reaching upto Additive Structure of a Number Ring.

Presented Talks

Cyclotomy: The Story of How Gauss Narrowly Missed Becoming a Philologist
Math Club, ISI Bangalore

August, 2025

Camps & Conferences

Madhava Nurture Camp

8-13 July, 2024

Organizers: HBCSE, CMI

Participated in Madhava Nurture Camp at CMI in summer of 2024.

Activities

Math Club, ISI Bangalore

Bengaluru, Karnataka

August, 2024 – Present

- Organized various talks by students as well as professors which can be found here:
[Math Club, ISI Bangalore](#).

Instructor at Ramanujan School of Mathematics(RSM)

Kolkata, West Bengal

December, 2023 - Present

- Aided students of grades 11 and 12 in their preparation for the entrance examinations of two of India's leading mathematical institutes, Indian Statistical Institute(ISI) and Chennai Mathematical Institute(CMI).

Skills & Interest

Technical: Java, C, Python, Octave, LaTeX, R.

Language: English, Bengali, Hindi

Interests: Reading Books, Painting.